

Reinforcement Learning

Shivaram Kalyanakrishnan

Department of Computer Science and Engineering
Indian Institute of Technology Bombay

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Reinforcement Learning

1. Reinforcement learning problem
2. Technical formulation
3. Applications of reinforcement learning
4. Conclusion

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Clear End, Unclear Means

Objective of the RoboCup Federation:

“By the middle of the 21st century, a team of fully autonomous humanoid robot soccer players shall win a soccer game, complying with the official rules of FIFA, against the winner of the most recent World Cup.”

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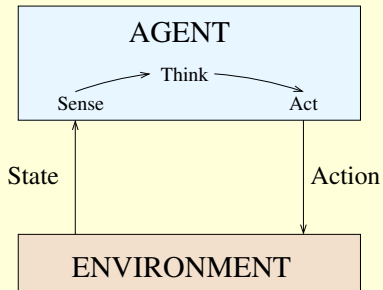
- Every robot is autonomous, executes a **computer program**.
How to **create** this program?
- Do we know how to **evaluate** a given program?
Yes! The program must maximise the probability of winning at soccer!

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Learning to Act Purposefully

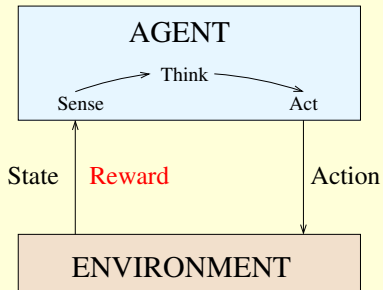
Answer: Reinforcement Learning (RL).

Learning to Act Purposefully



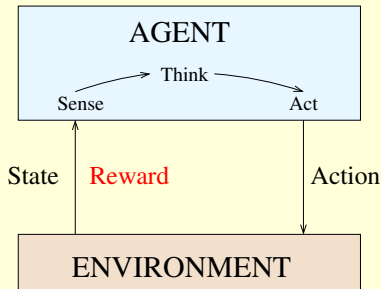
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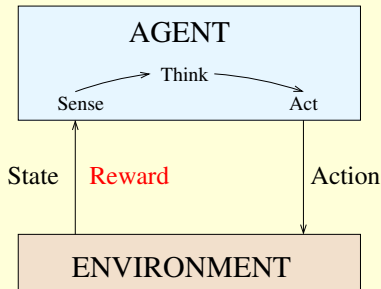
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Question: How must an agent in an *unknown* environment act so as to maximise its long-term reward?

Answer: Reinforcement Learning (RL).

Learning to Act Purposefully



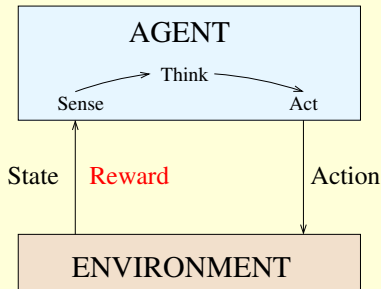
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[DeepMind Breakout video¹ (Mnih *et al.*, 2015)]

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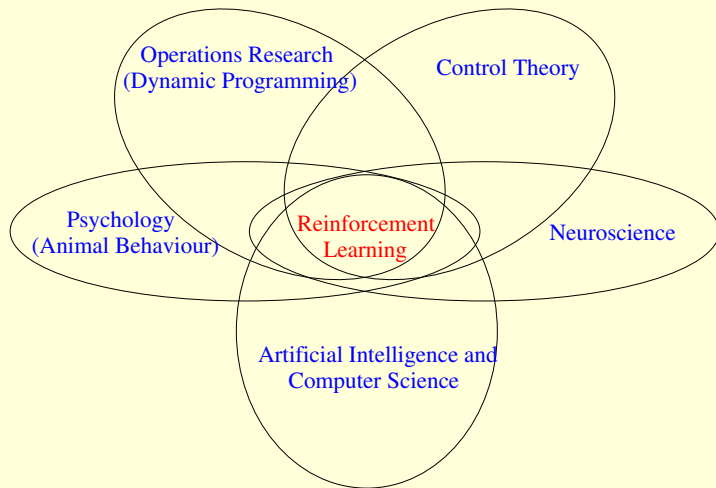
[Baby giraffe video²]

1. <https://www.youtube.com/watch?v=TmPfTpjtdgg>

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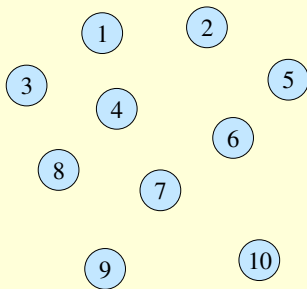
An Interdisciplinary Field



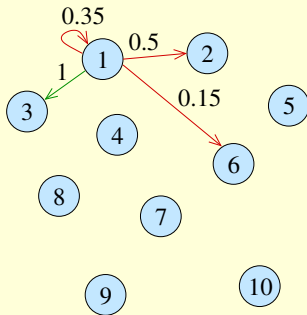
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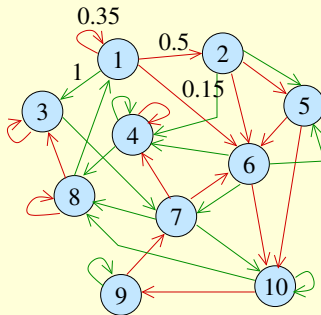
Markov Decision Problem (MDP)



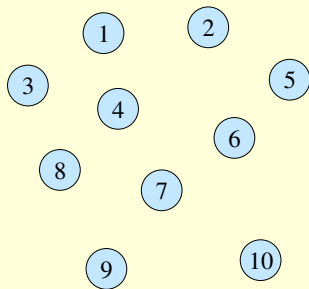
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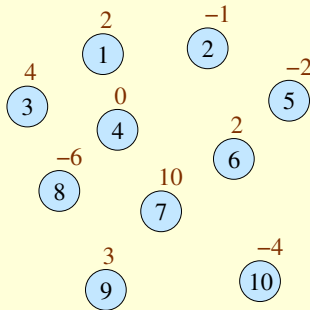
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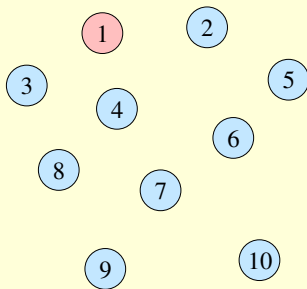
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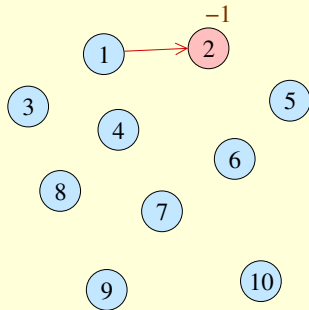
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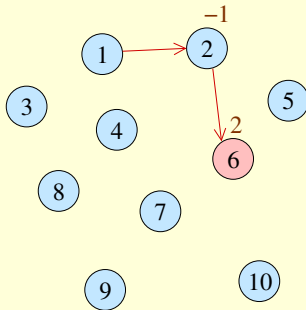


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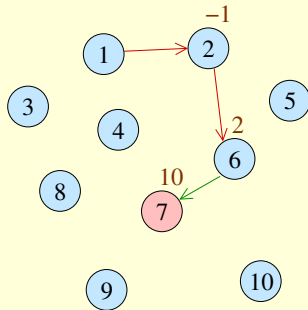
$$r_1 = -1.$$

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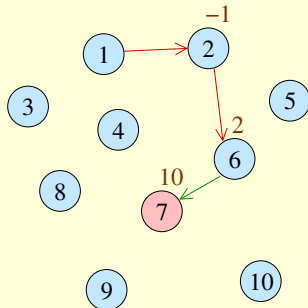
$$r_1 = -1. \quad r_2 = 2.$$

Markov Decision Problem (MDP)



$$r_1 = -1. \quad r_2 = 2. \quad r_3 = 10. \quad \dots$$

Markov Decision Problem (MDP)

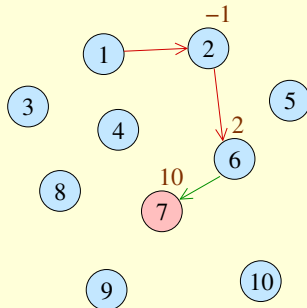


$$r_1 = -1. \quad r_2 = 2. \quad r_3 = 10. \quad \dots$$

How to take actions so as to maximise **expected long-term reward**

$$\mathbb{E}[r_1 + r_2 + r_3 + \dots]?$$

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How to take actions so as to maximise **expected long-term reward**

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[Note that there exists an (unknown) *optimal policy*.]

Q-Learning

- Keep a **running estimate** of the expected long-term reward obtained by taking each action from each state s , and acting *optimally* thereafter.

Q	red	green
1	-0.2	10
2	4.5	13
3	6	-8
4	0	0.2
5	-4.2	-4.2
6	1.2	1.6
7	10	6
8	4.8	9.9
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- Update these estimates based on **experience** (s_t, a_t, r_t, s_{t+1}):
$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha_t \{r_t + \max_a Q(s_{t+1}, a) - Q(s_t, a_t)\}.$$

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- Make sure to **explore** each action enough times.

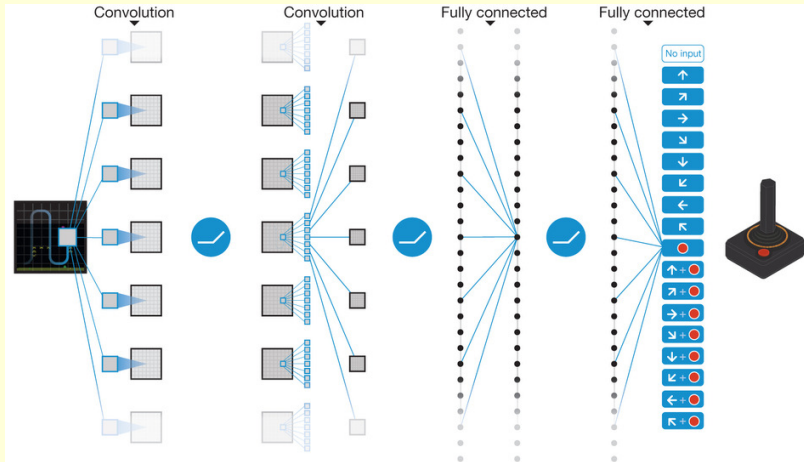
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- Act greedily based on the estimates (**exploit**) most of the time, but still
- Make sure to **explore** each action enough times.
- Q-learning converges and induces **optimal policy** (Watkins and Dayan, 1992)! ^{8/20}

Typical Neural Network-based Representation of Q



1. <http://www.nature.com/nature/journal/v518/n7540/carousel/nature14236-f1.jpg>

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Board Games

Backgammon



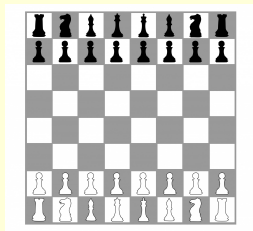
[1]

Go



[2]

Chess



[3]

References: Tesauro (1992), Silver *et al.* (2018).

1. <https://www.publicdomainpictures.net/pictures/60000/velka/backgammon.jpg>.

2. <https://www.publicdomainpictures.net/pictures/170000/velka/finished-go-game.jpg>.

3. <https://www.publicdomainpictures.net/pictures/80000/velka/chess-board-and-pieces.jpg>.

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Robotics and Control



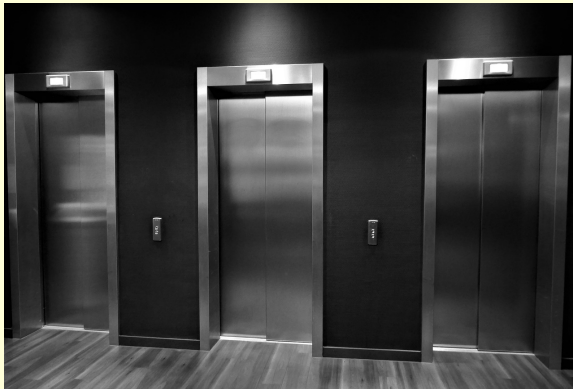
[1]

Reference: Ng *et al.* (2003).

1. <https://www.publicdomainpictures.net/pictures/20000/velka/police-helicopter-8712919948643Mk.jpg>.

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Planning and Scheduling



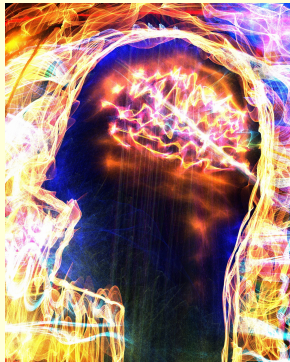
[1]

Reference: Crites and Barto (1995).

1. <https://www.needpix.com/photo/1029437/>.

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Adaptive treatment of epilepsy



[1]

- Reference: Guez *et al.* (2008).

1. <https://www.publicdomainpictures.net/pictures/140000/velka/brain-signals.jpg>.

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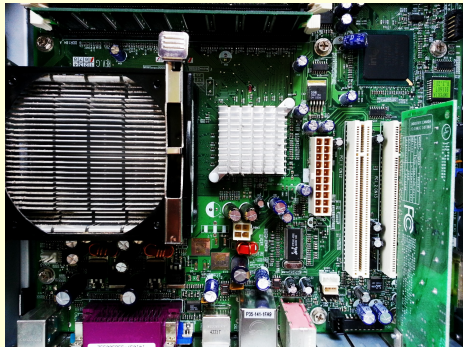
Stock trading



- Reference: Moody and Saffell (2001).

Computer Systems

Optimising a memory controller



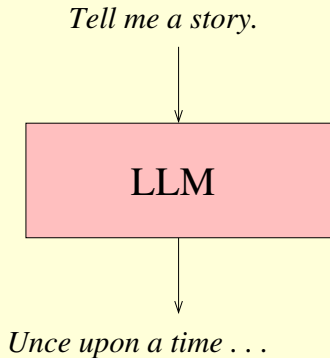
[1]

- Reference: İpek *et al.* (2008).

1. <https://www.publicdomainpictures.net/pictures/100000/velka/motherboard.jpg>.

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Large Language Models



Reference: Christiano *et al.* (2017).

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Summary

- How to make decision-making agents **autonomous**?
- RL idea: **Do not program behaviour!** Rather, specify goals.
- Rich history, at **confluence of several fields** of study, firm foundation.
- When applied using neural networks, can **learn** complex **non-linear** relationships in data.
- Demonstrated successes in a variety of **applications**.

Want to learn RL?

1. **Richard S. Sutton and Andrew G. Barto, 1998.** Reinforcement Learning: An Introduction. MIT Press, 1998.
2. **Leslie Pack Kaelbling, Michael L. Littman, and Andrew W. Moore, 1996.** Reinforcement Learning: A Survey. *Journal of Artificial Intelligence Research*, 4:237–285, 1996.
3. **NPTEL course** of B. Ravindran at IIT Madras.
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Thank you!